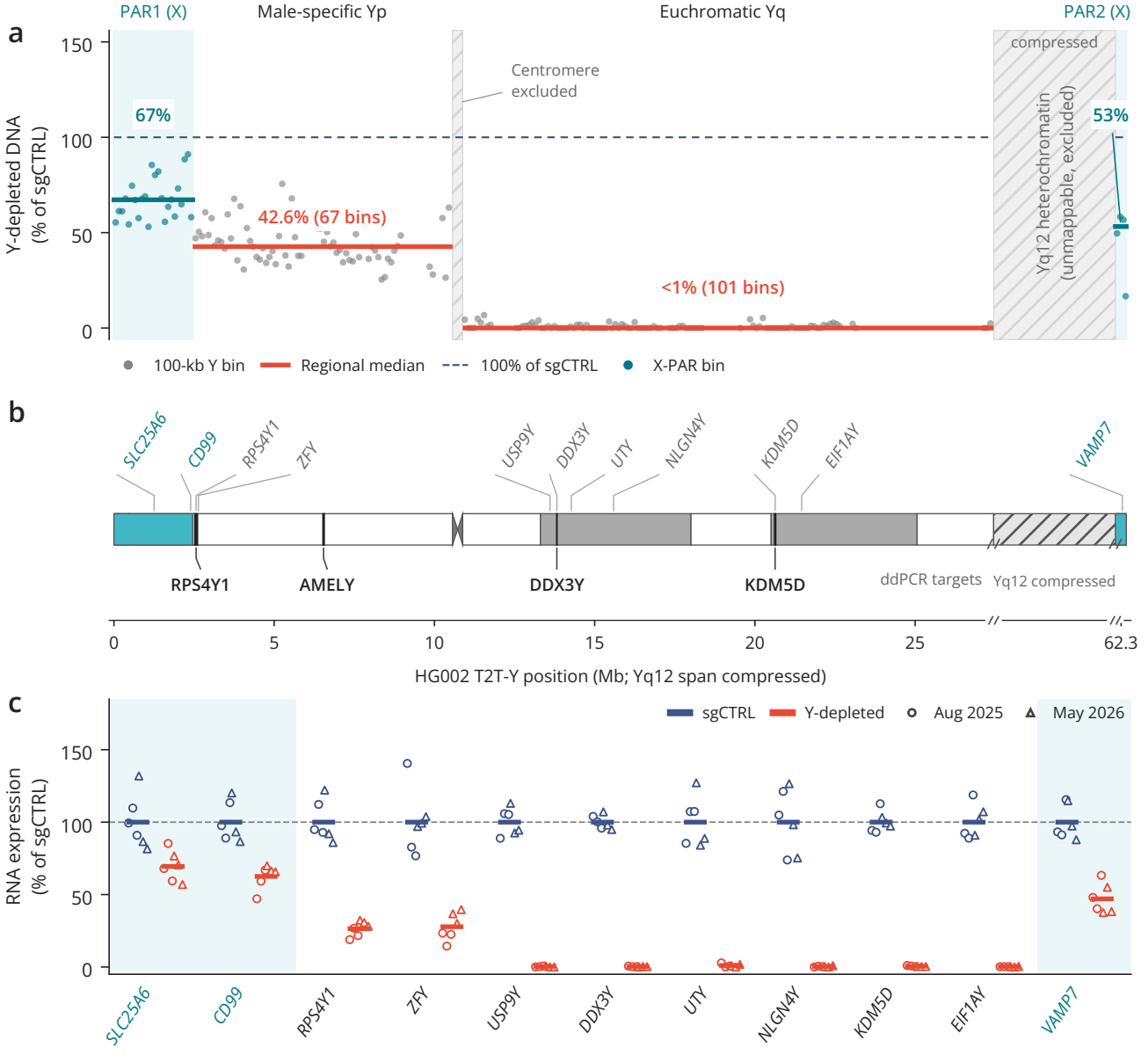
a
PAR1 (X)
Male-specific Yp
Euchromatic Yq
PAR2 (X)
Y-depleted DNA (% of sgCTRL)
150
100
50
0
67%
42.6% (67 bins)
Centromere excluded
<1% (101 bins)
compressed
Yq12 heterochromatin (unmappable, excluded)
53%
100-kb Y bin
Regional median
100% of sgCTRL
X-PAR bin
b
SLC25A6
CD99
RPS4Y1
ZFY
USP9Y
DDX3Y
UTY
NLGN4Y
KDM5D
EIF1AY
VAMP7
RPS4Y1
AMELY
DDX3Y
KDM5D
ddPCR targets
Yq12 compressed
0
5
10
15
20
25
62.3
HG002 T2T-Y position (Mb; Yq12 span compressed)
c
sgCTRL
Y-depleted
Aug 2025
May 2026
RNA expression (% of sgCTRL)
150
100
50
0
SLC25A6
CD99
RPS4Y1
ZFY
USP9Y
DDX3Y
UTY
NLGN4Y
KDM5D
EIF1AY
VAMP7